# Script

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Hello everyone. My presentation is about how computer networks may develop between now and 2035. My main question
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is how could artificial intelligence, quantum technology, blockchain and 6G
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can change future networks. I will explain what each technology does, where it fits into networking and what
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limitations we should keep in mind. And I will also show a small network experiment I ran on my Mac and present my proposed network of 2035.
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My main argument is that this technologies are more likely to extend today's internet than completely replace it.
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Before discussing future technologies, it helps to understand the problem networks already face.
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The first challenge is latency, which is the time it takes data to travel between endpoints. Latency can come from
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processing, queuing, transmission, and propagation. Processing delay is the time a device takes to examine a packet.
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Queuing delay happens when packets wait for the turn. Transmission delay is the time needed to place the packets bits
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onto a link. And propagation delay is the time the signal takes to travel through the medium. The second challenge
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is congestion. Congestion happens when more traffic competes for network resources than those resources can handle at that moment.
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Cues can grow which increases delay and packets may eventually be dropped. The third challenge is packet loss. A packet
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may fail to reach its destination because of congestion of faulty link, wireless interference or another problem.
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The fourth is throughput which is the amount of data successfully delivered over time. A connection may have a fast
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length but still achieve lower throughput if another part of the path is a bottleneck. A bottleneck is a part
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that limits the performance of the overall connection. These problems matter for future technologies because
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AI may help predict congestion and future wireless systems such as 6G may
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aim to support more demanding applications, but neither technology removes every physical or engineering limit.
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For a simple experiment, I used the terminal application on my Mac and ran the command shown here, ping, with four
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probes to google.com. The command sent four test packets and waited for replies. In my results, four packets were transmitted and four were received.
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So, the observed packet loss was 0%.
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The average round trip time or RTT was 37.538
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milliseconds. The minimum was about 19.536
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milliseconds and the maximum was about 76.287 milliseconds.
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Roundrip time measures the time for a probe to go to the destination and for its reply to return. The response times
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varied across the four probes. That variation could be related to changing queuing or network conditions. But this
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small experiment does not tell me the exact cause. It is important not to overstate the result. Four successful
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replies do not prove that the connection will never lose a packet.
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Also, ping does not directly measure the connection's maximum throughput and it doesn't identify a specific bottleneck.
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It is a useful basic observation, not a complete network diagnosis.
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In a future, network monitoring systems could collect measurements like delay and loss over longer periods. AI could
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then look for patterns and warn an operator about possible congestion. That is a possible use of
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AI, not something this four packet test demonstrates.
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This slide shows three ways AI can support network management. Predict, optimize, and detect. Force prediction
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means learning from traffic history and current measurements to estimate when
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traffic may increase or congestion may happen. For example, a network might see a regular increase in demand at a particular time of a day.
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Second, optimization means using those estimates to recommend better routes or
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allocate available bandwidth and computing resources. A system might move some traffic to a less busy path if that
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path is available and the change is safe.
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Third, detection means identifying unusual traffic, equipment faults, or behavior that may indicate an attack.
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AI can help operators find patterns that would be difficult to spot manually. AI
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can operate in different places a device or edge system may make a quick local
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decision while a cloud platform can analyze larger amounts of data.
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A network controller may combine information from different routers and links. There are important risks. AI
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depends on the quality or representativeness of its data.
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Incorrect predictions can lead to poor decisions. Attackers may also try to manipulate the data and an automatic
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change can disrupt service if it is unsafe.
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That is why monitoring, testing, human oversight and a way to reverse changes remain important.
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My view is that AI may have a broad impact because it can improve existing networks without requiring every router or protocol to be replaced.
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Let's see quantum networking. So quantum networking is different from ordinary networking because it involves
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communicating quantum quantum information rather than only classical bits. Some proposed systems use photons
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to carry quantum states between nodes. A future quantum network could connect quantum devices and support tasks such
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as distributing entanglement between locations.
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Quantum repeaters are being researched to help extend quantum connections over longer distances. But building practical
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repeaters and maintaining fragile quantum states are difficult challenges.
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The diagram on this slide is a simplified conceptual link between node A and node B. It is not meant to show a normal internet route.
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A quantum link is specialized and would not replace the ordinary IP path used for everyday websites, messaging or
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video calls. Quantum networking is still experimental compared with today's commercial internet. The important
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distinction is that it may add new capabilities for particular applications, but it does not mean all
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internet traffic will suddenly travel through quantum links.
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Let's see postquantum security. So postquantum cryptography or PQC addresses a different problem from quantum networking.
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Some public key public key cryptographic systems used today could be threatened by a sufficiently capable quantum
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computer. The concern is not that every current encryption method immediately stops working, but that organizations
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need to prepare for future attacks against certain widely used public key methods.
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PQC used uh cryptographic algorithms designed to resist attacks
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from both classical and quantum computers. In 2024, NISC published three
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key standards and organizations still need to plan and carry out migrations because cryptography is built into software
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devices, certifications and network equipment.
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The key distinction is that PQC protects classical digital communication using new cryptographic algorithms. Quantum
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networking is about communicating quantum information. They are related to the broader development of quantum
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technology, but they are not the same thing.
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For a network of 2035, I expect PQC to be important because protecting ordinary
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data and systems is a practical security need even if large scale quantum networks remain experimental.
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Let's see blockchain in networking.
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Blockchain is a way for multiple participants to maintain a shared network in which changes can be varied
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according to agreed rules. For networking, the useful question is not how to use cryptocurrency, but whether a
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shared record solves a real coordination or trust problem. Possible examples
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include recording device identities, tracking changes to access permissions, and maintaining security audit logs that
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multiple organizations can verify. This could be useful in some internet of things environments where devices or
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services are managed by different parties.
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However, blockchain has tradeoffs. Cons can add delay. Records can consume
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storage and privacy needs careful design. The system can also become more complicated to operate. Blockchain does
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not automatically make a device or application secure. The surrounding identity, key management, software, and
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access controls still matter. If one trusted organization already controls the system, a conventional database may
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be faster, cheaper, and easier to maintain. Blockchain is most useful when multiple parties need to coordinate
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records without relying entirely on one party to maintain the only authorative copy.
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Let's see. 6G, the next mobile generation. The additional technology I selected is 6G. The next generation of
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mobile networking after 5G. The expected direction includes future mobile
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connectivity, more integration with AI and support for connected devices and
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demanding services. Research also explores combining communication with
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sensing and using new tech new new frequency ranges.
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These are goals and research directions not a promise that every 6G system will provide every capability. These are
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challenges. New radio technologies need testing and standards. Deployment may require expensive infrastructure and
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energy use. Coverage, security and affordability all matter.
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Very high frequencies can offer wide bandwidth but may have shorter range and be more affected by obstacles.
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The slide labels 6G as emerging and in the standardization process rather than a fully deployed commercial generation.
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That matters because we should distinguish an intended capability from a technology that has already been
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proven at large scale. In my proposed architecture, 6G is one possible excess
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technology. It connects devices to the network. But the higher level IP, transport, application and security functions remain important.
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This slide connects the future technologies to the protocol layers. At the application layer, users interact
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with services. AI services and blockchain applications can operate here. At the transport layer, TCP or UDP

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provides communication between applications.
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At the network layer, IP addressing and routing move packets across networks. The link layer carries
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data across a particular network link such as Wi-Fi or a mobile radio connection. The physical layer carries
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the actual signals through fiber, copper or radio. Encapsulation is the process of wrapping data
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as it moves down the stack. a message from an application which is carried in a transport segment which is carried in
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an IP packet which is carried in a link layer frame and finally represented as
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bits or signals on the medium. The exact terminology can vary by protocol but the
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basic idea is that each layer adds information needed for its job. The important point is that AI, blockchain,
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6G and PQC do not simply erase the protocol stack.
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They may add services, change how a layer is implemented or improve security and management. The layers, they'll
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provide a useful way to organize responsibilities and troubleshoot problems.
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Let's see my network of 2035. This is my proposed network of 2035 architecture.
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It is a conceptual design, not a claim that all these technologies already work together in one deployed system. The
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main classical data path begins with end devices. They connect through an access network which could use Wi-Fi, 5G, 6G or another connection.
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From there, data travels through the IP network using routers and transport protocols such as TCP or UDP.
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Applications may run at an edge location close to users or in a cloud data center. The AI network controller is
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shown separately because it has a management role. It can collect telemetry, monitor traffic, predict
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congestion and re and recommend or apply routing and resource changes. The dashed
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control path is distinct from the normal path that carries user packets. Cyber security applies across the architecture. Encryption protects data.
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Authentication helps verify identities and thread detection looks for suspicious behavior.
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PQC can be introduced into suitable cryptographic systems as organizations migrate to quantum resistant algorithms.
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Blockchain is shown as a separate shared record function. It may store selected device identity information access changes or audit locks.
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It does not need to carry every packet and it would be only be included where shared networks, shared records provide
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a clear benefit. The quantum pathway is also separate. It represents a specialized experimental link that could
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connect quantum capable nodes using photons or future quantum repeaters.
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It is not the road that ordinary IP packets take. The diagram therefore has several distinct roles. Classical data
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transport, AI management, security across the path, shared records when useful, and a specialized quantum network.
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My prediction is that most of the everyday network will remain based on conventional internet protocols with
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newer technologies added where they solve specific problems.
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This matrix compares the technologies by purpose and maturity. AI networking focuses on management and optimization.
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Some AI assisted tools are already used while more autonomous or reliable networkwide control continues to develop.
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Quantum networking carries quantum information and remains experimental.
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Postquantum security protects classical cryptography. Standards have been published but migration is ongoing.
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Blockchain supports shared and verif ver verifiable records but its usefulness depends on the application.
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6G is the next mobile generation and remains in an emerging and standardization phase. This technologies
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should not be treated as competitors doing the same job. AI help helps operator network 6G 6G changes future
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wireless access capabilities. PQC helps protect classical communications against future cryptographic threats. Blockchain
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can coordinate shared records and quantum networking may support specialized quantum applications. My
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prediction is that AI native networking may have the broadest everyday impact over the next decade. The reason is that
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AI can be added to existing networks for traffic prediction, fault detection and resource management. It may improve the
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performance of infrastructure people already use. This is a prediction, not a certaintity. The impact will depend on
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reliability, security, costs, and whether AI systems can be tested and controlled safely. The best future
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network will likely combine technologies selectively rather than use every new technology everywhere.
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Let's conclude this video. So to conclude, the future internet is more likely to evolve than to be replaced. AI
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may help networks predict problems and manage resources. 6G may expand future
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wireless connectivity. Post quantum cryptography is important for protecting classical digital communication. As
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quantum computing develops, blockchain may help maintain shared records in settings where several parties need to trust the same history.
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Quantum networking could enable specialized new capabilities, but large scale systems remain experimental. The
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main lesson from this project is to evaluate each technology by the problem it solves. whether it fits in the architecture.